

Grokking Under Federated Averaging: Robustness and Failure Modes

1 Setup

- **Task:** modular addition $(a + b) \bmod 97$, training fraction $\alpha=0.5$
- **Model:** GrokNet — 2-layer MLP, no biases, quadratic activation, mean-field init, $N=128$, MSE loss
- **Centralised baseline:** full-batch GD, lr=50, wd=0
- **Federated:** Flower simulation, K clients, 2000 rounds, 5 local epochs (defaults)
- **Partition schemes:** IID, operand-based ($a \bmod K = k$), target-based ($(a+b) \bmod K = k$)

2 Experiment 1: Federated vs. Centralised Baseline

- FedAvg (IID, $K=5$, full participation) reproduces the centralised grokking trajectory step-for-step (Figure 1)
- Both reach 100% test accuracy at ~ 8200 gradient steps with identical IPR dynamics ($0.02 \rightarrow 0.34$)
- All three partition strategies (IID, operand, target) produce indistinguishable curves at $K=5$

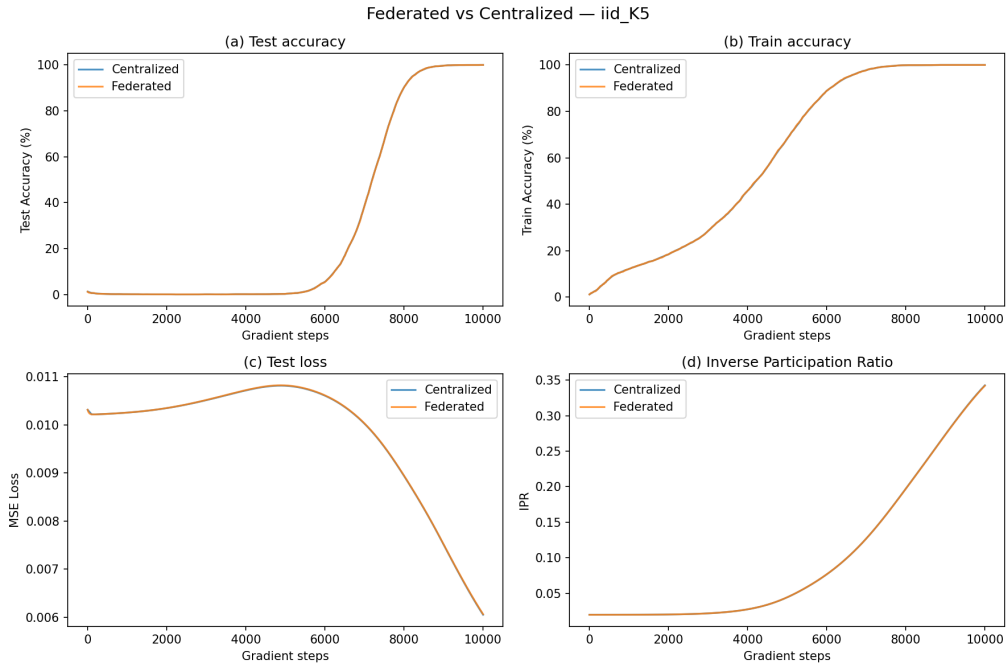


Figure 1: Federated (IID, $K=5$) vs. centralised training. Trajectories are effectively identical.

3 Experiment 2: Stress-Testing Federated Grokking

Three axes probed to find where federated grokking breaks.

3.1 Client Scaling (IID, full participation)

Table 1: Client scaling. All configurations grok to 100%.

K	Samples/client	Grok step	Final test acc	Final IPR
5	940	8215	100.0%	0.342
10	470	8220	100.0%	0.342
20	235	8240	100.0%	0.340
50	94	8280	100.0%	0.332
97	48	8430	100.0%	0.311

- $K=97$ (48 samples/client) still groks to 100%
- Grok step increases only 2.6% from $K=5$ to $K=97$
- Grokking is immune to client count under IID

3.2 Partial Participation ($K=20$, target partition)

Table 2: Partial participation under non-IID target partition ($K=20$).

Fraction	Clients/round	Grok step	Final test acc	Final IPR
1.0	20	9585	98.5%	0.214
0.8	16	9600	97.5%	0.218
0.6	12	9725	96.2%	0.220
0.4	8	9900	93.9%	0.222
0.2	4	9720	94.5%	0.242

- **Only axis that degrades grokking**
- Non-IID target + low participation ($f_t=0.2$) caps accuracy at $\sim 94\%$ vs. 100% under IID
- Cause: not all target residue classes represented each round $\Rightarrow$ systematic under-representation
- Grokking still occurs — accuracy jumps from $\sim 1\%$ to $\sim 94\%$ at roughly the same step

3.3 Extreme Local Epochs ($K=5$, IID)

Table 3: Extreme local epochs ($K=5$, IID, full participation).

Local epochs	Rounds	Grok step	Final test acc	Final IPR
50	400	8300	100.0%	0.457
100	200	8400	100.0%	0.446
200	100	9000	100.0%	0.426

- $le=200$ (only 100 communication rounds) still groks to 100%
- Higher local epochs produce *higher* IPR (0.46 vs. 0.34 at $le=5$)
- Extended local training strengthens periodic Fourier features rather than harming them

Breaking Point: Where Does Federated Grokking Fail?

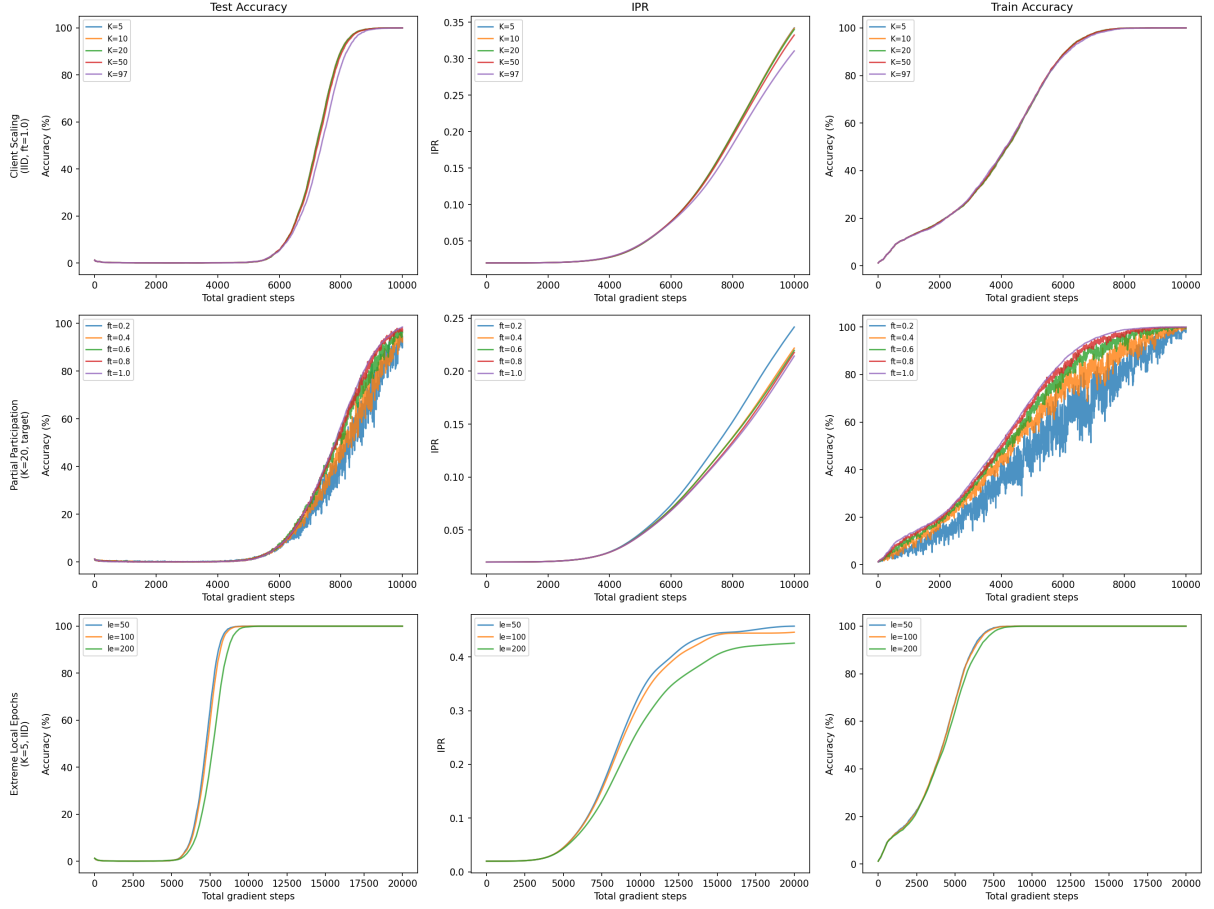


Figure 2: Breaking point: 3×3 grid. Rows: client scaling, partial participation, extreme local epochs. Columns: test accuracy, IPR, train accuracy.

4 Experiment 3: FedProx Sweep

- FedProx adds proximal term $\frac{\mu}{2} \|w - w^{\text{global}}\|^2$ to each client’s local objective
- Sweep: $\mu \in \{0, 0.001, 0.01, 0.1, 1.0, 10.0\}$, $K=5$, IID, full participation

Table 4: FedProx sweep. Sharp transition between $\mu=0.001$ (groks) and $\mu=0.01$ (fails).

μ	Label	Final test acc	Grok step	$\ W_1\ _F$	Status
0	FedAvg	100.0%	8215	23.80	Groks
0.001	FedProx	99.7%	9075	22.39	Groks (10% delayed)
0.01	FedProx	0.1%	—	15.14	Weight growth suppressed
0.1	FedProx	1.0%	—	NaN	Diverges
1.0	FedProx	1.0%	—	NaN	Explodes (round 2)
10.0	FedProx	1.0%	—	NaN	Explodes (round 1)

- **Sharp phase transition** between $\mu=0.001$ and $\mu=0.01$
- $\mu=0.001$: groks with 10% delay (step 9075 vs. 8215), slightly lower weight norm (22.4 vs. 23.8)
- $\mu=0.01$: weight norm capped at 15.1 (vs. 23.8), only 44% train accuracy, no grokking
- $\mu \geq 0.1$: proximal gradient $\mu(w - w^{\text{global}}) \times \text{large lr (50)} \Rightarrow$ immediate numerical diver-

gence

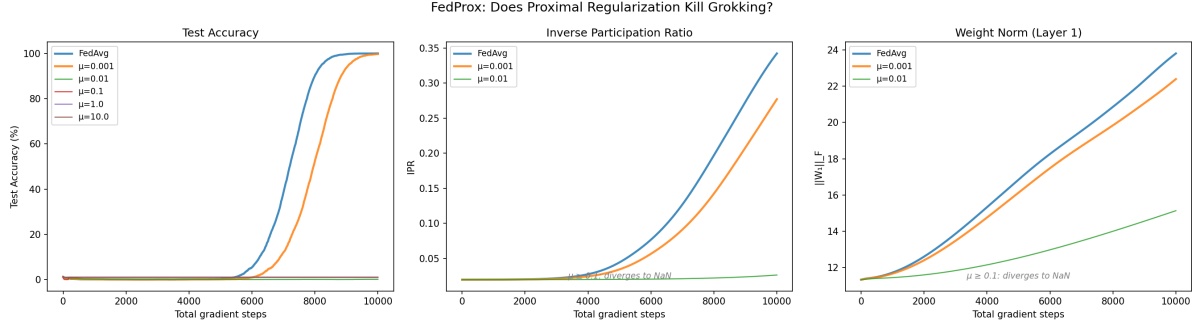


Figure 3: FedProx sweep. Left: test accuracy. Centre/Right: IPR and weight norm ($\mu \leq 0.01$ only; $\mu \geq 0.1$ diverges).

5 Discussion

Grokking is a property of the loss landscape, not the optimisation path.

- FedAvg’s noisy, distributed optimisation does not prevent the phase transition
- Tested: up to 97 clients, as few as 4 clients/round, up to 200 local epochs
- The grokking mechanism (periodic Fourier mode learning via weight norm growth) depends on loss landscape geometry, which is preserved regardless of how gradient steps are distributed

The proximal term directly opposes grokking.

- Grokking requires weight norms to grow from ~ 11 to ~ 24 (“lazy” $\rightarrow$ “rich” regime)
- FedProx’s $\frac{\mu}{2} \|w - w^{\text{global}}\|^2$ acts as additional weight decay, anchoring parameters near the global model
- Even $\mu=0.01$ suppresses growth below the critical threshold ($\|W_1\|_F \approx 15$ vs. 24)
- **Fundamental tension:** the regularisation that makes FedProx effective for heterogeneous FL (reducing client drift) directly conflicts with the weight norm growth that enables grokking

Partial participation is the only weak point.

- Only non-IID target partition + low participation ($f_t=0.2$, $K=20$) degrades accuracy (to $\sim 94\%$)
- Not a failure of grokking itself — the sharp transition still occurs
- Information-theoretic limitation: with 4 of 20 clients/round, some target residue classes are systematically under-represented